

TAU-MULTILINGUAL: BENCHMARKING VOICE AGENTS ACROSS LANGUAGES

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ABSTRACT

English-only benchmarks expose only a narrow slice of voice-agent behavior. We introduce τ -Multilingual, extending τ -Voice to Spanish, Brazilian Portuguese, Hindi, Korean, and Mandarin with native-speaker review and evaluation of generated language and spoken output. Across 4,500 full-duplex calls and five voice configurations, Spanish, Portuguese, and Hindi remain within 3.2 task-completion points of English, but Korean and Mandarin fall by 16.8 and 11.3 points. The failure modes also vary: Korean systems miss more responses, Mandarin systems interrupt more often, and both struggle with tools and entities. Grok leads task completion but scores lowest on generation quality, motivating separate task, interaction, and generation reporting. We release language packs, validated judges, and tools for community-built multilingual voice-agent evaluation.

Index Terms— multilingual voice agents, spoken dialogue evaluation, task-oriented dialogue, full-duplex interaction, user simulation

1. INTRODUCTION

Voice agents must serve a multilingual world: English is spoken by only 19% of the world’s population [1], yet leading end-to-end voice-agent benchmarks evaluate English alone [2, 3, 4]. Multilingual evaluation must therefore capture task completion, conversational behavior, linguistic naturalness, and speech fidelity.

We introduce τ -Multilingual, extending τ -Voice [2] with three contributions. First, *LanguageFactory* provides a reviewed workflow for native-speaker-authored simulators, localizing caller personas, speech rules, and identities while preserving the underlying tasks. Second, we add human-validated evaluation of linguistic naturalness and speech fidelity alongside task completion and interaction measures. Third, we evaluate five voice configurations and two text systems across six languages and three domains.

We release the codebase [5] so the community can build callers reflecting local linguistic and cultural norms, rather than merely translated English personas, alongside language-specific audits of English-centric agent behavior.

2. RELATED WORK

τ -Voice brings the database-grounded tasks of τ -bench and τ^2 -Bench [13, 14] to full-duplex speech, evaluating task completion and conversational dynamics but not generation quality [2]. It, EVA-Bench, and FDB-v3 are English-only [3, 4]. M3-DuplexBench covers English and Japanese turn-taking without

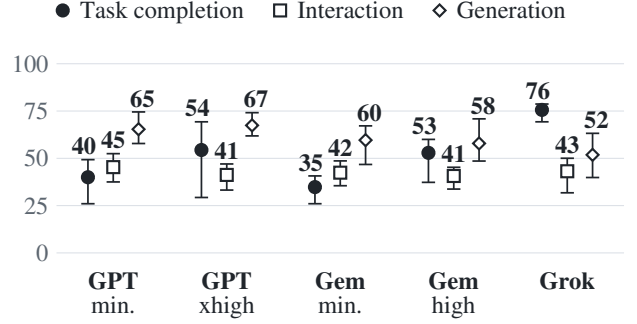


Fig. 1. System means across languages with worst-to-best language whiskers; Generation excludes English.

Table 1. Related benchmarks. A check mark denotes direct evaluation and a tilde partial coverage. Spk.: spoken; FD: full-duplex runtime; Mult.: multilingual; Int.: interaction-quality evaluation; Lang. quality: naturalness and language use.

Work	Spk.	FD	Mult.	Task	Int.	Lang. quality
<i>Voice-agent benchmarks</i>						
τ -Voice [2]	✓	✓	—	✓	✓	—
FDB-v3 [4]	✓	✓	—	✓	✓	—
EVA-Bench [3]	✓	~	—	✓	✓	~
M3-DuplexBench [6]	✓	✓	✓	—	✓	—
<i>Multilingual speech and dialogue</i>						
VoiceAgentBench [7]	✓	—	✓	~	—	—
Speech-MASSIVE [8]	✓	—	✓	—	—	—
X-RiSAWOZ [9]	—	—	✓	—	—	—
GlobalWoZ [10]	—	—	✓	—	—	—
<i>Language-quality evaluation</i>						
MQM [11]	—	—	✓	—	—	✓
MENLO [12]	—	—	✓	—	—	✓
τ-Multilingual	✓	✓	✓	✓	✓	✓

executed tool tasks [6]; VoiceAgentBench evaluates multilingual spoken tool use at the query/workflow level [7]; and Speech-MASSIVE evaluates intent and slot prediction from recorded speech [8]. X-RiSAWOZ and GlobalWoZ provide multilingual text dialogue with human-verified or locale-native content [9, 10]; MQM and MENLO motivate factorized, human-reviewed language evaluation [11, 12]. τ -Multilingual combines live voice calls, tools, environment execution, and naturalness audits (Table 1).

3. METHODOLOGY

3.1. Matched Multilingual Benchmark

τ -Voice supplies English task frames, agent policies and tools, user scenarios, and a simulated caller [2]. Its full-duplex runtime advances in 200-ms ticks; binary rewards check task-specified environment, action, and communication criteria. Each language pack localizes the caller instructions, persona, voice, identity entities, and evaluation rules while preserving the underlying tasks and scoring. The same specification drives both voice and matched text conditions, enabling comparisons by domain, task, system, and reasoning setting. LanguageFactory stores this specification as a typed, versioned artifact; Table 2 illustrates the Hindi pack. Cross-language differences therefore compare complete localized systems rather than an isolated causal effect of language.

Table 2. LanguageFactory components illustrated with the Hindi pack. Entity localization maps the same source identity to locale-specific backend values without changing the task.

Language component	Hindi (India)
Personas	Rishika uses fast Hinglish; Imran uses patient, Urdu-inflected Hindi.
Speech rules	Spell with “R for Rajdhani”; say <i>at</i> and <i>dot</i> in email addresses.
Task arms and entity localization	Allison Reeves → Neha Gupta; phone: 99400 97181.
Audit factors	Customer address (<i>aap</i> vs. <i>tum/tu</i>) and agreement with <i>aap</i> .
Review and freeze	Pin the Rishika and Imran voices; freeze preset <code>multilingual-v1.hindi</code> .

Native speakers review each pack’s language rules, voices, and pilot calls. Automated checks enforce schema, task parity, and persona assignment before release.

3.2. Evaluation Cohorts

The benchmark contains 900 localized task instances: 50 tasks in each of three domains, instantiated in six languages. Five voice configurations yield 4,500 primary calls: GPT (OpenAI `gpt-realtime-2`) at minimal and xhigh reasoning, Gemini 3.1 Flash Live Preview at minimal and high, and Grok Voice Think Fast 1.0 [15, 16, 17]. The cascaded caller uses `gpt-5.5` xhigh for text and Eleven v3 for speech [18, 19]. For task-completion significance tests only, the four GPT and Gemini configurations include one additional trial. We use the short labels GPT, Gemini, and Grok below.

The matched half-duplex text control evaluates the same 900 localized task instances with two systems (1,800 calls): GPT-5.5 xhigh and Gemini 3.1 Pro Preview high [18, 20]. Because both the model and interaction regime change, this is an end-to-end control rather than a pure acoustic intervention.

The ablation evaluates the same 30 retail tasks in Hindi and Mandarin with GPT xhigh and Gemini high. Against the localized, romanized baseline, *Entity* substitutes source-English identity values, while *Entity* + *DB* adds native-script values and matched

spell-out, with all else fixed.

3.3. Evaluation Measures

Task completion is the call-level environment-reward pass rate. We introduce two complementary quality scores. *Interaction* is a call-level composite: 100 minus the equal-weight mean of non-response (missed/late replies or failure to yield), inappropriate interruption (talking over the caller), selectivity error (reacting to backchannels, vocal tics, or non-directed speech), monologues over 45 seconds, and validated tool misuse (incorrect arguments or agent-caused errors). *Generation quality* (Generation) is the share of aligned agent utterances passing both linguistic naturalness and speech fidelity checks. These composites are compact representations only; their intermediate values have no calibrated interpretation. Naturalness flags unnatural phrasing. Speech fidelity uses a multimodal Gemini 3.1 Pro Preview judge [20] to compare each scored agent-audio clip with its expected text. A severity- ≥ 2 delivery finding or Mandarin tone-induced meaning flip is a failure. Language-specific diagnostics remain separate. English is excluded only from Generation because our Naturalness rubric targets translationese and native phrasing in the five localized languages.

We compare systems within languages and estimate matched performance gaps relative to English for Task completion and Interaction. We do not rank languages: each is a complete localized system, and naturalness is not cross-language calibrated.

3.4. Human Validation

LLM-based judges enter results only when precision and recall exceed .75, F1 exceeds .80, and Cohen’s $\kappa > .60$. For naturalness and language-specific diagnostics, labels require consensus from two native-language annotators. Validation covers tool use, approximately 60 examples per language for naturalness and speech fidelity, and at least 30 per language-specific diagnostic; failing measures remain archived.

Separately, reviewers audit 30 completed calls per language, balanced across domains and systems. They attribute the first critical error to the agent, simulator, or infrastructure and score eight caller-quality dimensions—including voice/prosody quality, turn-taking naturalness, and behavioral plausibility—on a four-point scale (see the released audit rubric [5]).

3.5. Statistical Analysis

Task-completion inference averages the two trials per task for the four GPT/Gemini configurations, then compares each language with English across 150 matched domain–task clusters using paired permutation tests over five contrasts. We report paired BCa 95% intervals from 10,000 resamples. Grok has one trial and appears only in the descriptive estimates in Figure 2(a).

Interaction and Generation use 100,000 paired cluster-level label-swap permutations. Interaction compares each language with English after jointly swapping the five systems and reaggregating component-specific failure and opportunity counts. Generation compares all ten system pairs after jointly swapping the

	(a) Task completion						(b) Interaction						(c) Generation				
	EN	ES	PT	HI	KO	ZH	EN	ES	PT	HI	KO	ZH	ES	PT	HI	KO	ZH
GPT min.	45	46	49	43	26	30	52	46	46	48	38	42	64	69	58	61	75
GPT xhigh	65	58	69	61	29	44	47	40	43	45	33	39	68	74	62	62	71
Gem min.	41	37	38	39	26	28	49	44	43	48	36	36	58	67	47	66	60
Gem high	51	60	59	60	37	50	45	42	45	42	36	34	61	71	49	57	52
Grok	76	78	79	76	75	69	50	47	47	44	39	32	50	60	40	47	63
Overall	56	56	59	56	39	44	49	44	45	45	36	36	60	68	51	59	64

Fig. 2. Headline outcomes by system and language. Higher is better; color scales normalize within panel, outlines mark language-best systems, Overall averages the five configurations, and Generation excludes English.

Metric	GPT min.	GPT xhigh	Gem min.	Gem high	Grok	Avg.
Task failure _c (%)	60	46	65	47	24	48
Non-response _c (%)	9	28	47	77	10	34
Interruption _c (%)	80	90	74	79	90	83
Selectivity error _c (%)	97	96	84	77	97	90
Monologue _c (%)	18	18	7	11	37	18
Tool misuse _c (%)	70	62	77	53	51	62
Naturalness failure _u (%)	32	30	37	38	44	36
Speech-fidelity failure _u (%)	7	7	8	9	21	10
Latency _r (s)	1.12	1.42	1.36	1.82	1.20	1.38

Fig. 3. Main-metric breakdown across languages by system; lower is better. Bold numbers mark row minima. Subscripts c/u/r denote call/utterance/response; units are in parentheses.

five localized calls and reaggregating aligned utterances. All reported confirmatory p -values use Holm correction within each family: five language contrasts for Task completion and Interaction, and ten system pairs for Generation. The retail ablations are descriptive.

4. RESULTS

Figures 2 and 3 summarize task completion, Interaction, Generation, and their component measures. We then examine localization failures and model choice.

4.1. Task Completion

Task completion is broadly preserved in Spanish, Portuguese, and Hindi but not in Korean and Mandarin (Figure 2(a)). The confirmatory English-relative losses for Korean and Mandarin are 20.7 and 13.3 points (both Holm-adjusted $p < .00005$); the other three differences are not significant. The domain pattern is similarly misaligned: telecom is hardest by task completion (44.1%) yet yields the highest Generation (63.2%).

The voice–text gap favors pooled text systems in every language by 16.8–41.5 points. The smallest gap is Spanish; the largest is Korean, where text reaches 80.3% versus 38.8% voice. Because model and turn-taking also differ, this is an end-to-end voice-system gap rather than a purely acoustic effect.

4.2. Interaction

Interaction shows the same broad split but a different inferential result: all five English-relative declines are significant ($p \leq .00055$), with substantially larger effects for Korean and Mandarin. The Korean gap concentrates in non-response (46.1% versus 27.8% in English) and tool-use failure (84.9% versus 49.8%). Mandarin instead combines elevated interruption (91.7% versus 78.0%) with tool-use failure (78.8%). Selectivity failure is high in every language (86.3–93.5%), so it is a broad system weakness rather than the main explanation of the language gap. Spanish and Portuguese show more monologuing than English, but not the Korean/Mandarin non-response and tool-use pattern. Similar aggregate Interaction losses thus arise from different operational failures and imply different remedies.

4.3. Generation Quality

Generation produces a different model ranking: GPT xhigh leads Spanish, Portuguese, and Hindi; Gemini minimal leads Korean; and GPT minimal leads Mandarin (Figure 2(c)).

In their naturalness and cultural-grounding audits, native reviewers found recurring literal or interface-like language: Spanish *user id*, Portuguese “nenhum voo ainda foi voador” (no flight has yet been flown), Hindi “explicit yes,” Korean spoken tool names such as *check_sim.status*, and Mandarin *toggle data*. Targeted audits capture different conventions: mixed formal and informal address in Portuguese; gendered self-reference and omitted *-ji* in Hindi; missing subject honorification in Korean; and unsuitable titles and missing sentence-final particles in Mandarin. These diagnostics contextualize Naturalness but are not components of Generation. GPT is more natural in Hindi than Gemini and Grok (35.1–40.6% versus 49.7–55.5% failure), yet GPT minimal has the most gender errors (49.3%). In Mandarin, GPT has nearly all modal-particle failures (40.0–59.1% versus 0–2.0%).

A related, separate diagnostic is gender agreement. Agent gender was inadvertently absent from the prompt despite using female voices, yielding inconsistent self-grounding. Hindi agents default to masculine self-reference (49.5% of calls versus 5.3% with any feminine form). The overall gender-agreement failure rates—27.2% in Hindi and 29.6% in Portuguese—combine self-reference, noun/anaphora agreement, and caller-directed errors. Disclosure changes the mix but does not reliably reduce the total:

in 30-call-per-language ablations, failures rise from 30.0% to 40.0% in Hindi and fall from 30.7% to 20.0% in Portuguese.

Speech fidelity reveals an all-language provider pattern: Grok’s cleanliness is 79.7%, versus 91.4–93.9% for the other systems. Across 71,413 utterances, the most common material failures are punctuation or formatting errors (e.g., speaking punctuation aloud; 3,529 utterances), mispronunciations (1,362), and number, date, or currency errors (1,068).

4.4. Localization and Entity Handling

The large Korean and Mandarin task losses concentrate in authentication: exact lookup succeeds in 81–85% of English calls, but only 17–55% of Korean and 37–58% of Mandarin calls across repeated GPT/Gemini systems. Localized caller prompts provide the correct family-first name, yet agents often map the spoken parts into a given-first lookup. In Mandarin, this can break exact lookup even when callers provide the correct romanized spelling and the agent prompt requires all tool calls, arguments, and structured outputs in English; native-character explanations add a separate cross-script mismatch. Telecom’s balanced split shows an exploratory crossover: Korean task completion is 32.0% for name+DOB versus 35.2% for phone authentication, whereas Mandarin is 40.8% versus 33.6%; method remains bundled with task.

Table 3. Retail entity ablations: task completion (%) on 30 tasks.

Lang.	System	Entity		Entity + DB	
		Localized	Source	Romanized	Native
Hindi	GPT xhigh	63.3	63.3	63.3	36.7
	Gemini high	60.0	66.7	60.0	60.0
Mandarin	GPT xhigh	30.0	46.7	30.0	46.7
	Gemini high	43.3	53.3	43.3	60.0

In the localization ablations, source-English entities match or outperform localized ones in every cell of Table 3, improving Mandarin by 16.7 points for GPT and 10.0 for Gemini. Replacing romanized database values with native-script names improves Mandarin by 16.7 points for both systems. Hindi differs: native script leaves Gemini unchanged and lowers GPT by 26.7 points. These 30-task cells are exploratory; exact authentication does not ensure task completion.

4.5. Model Choice Across Outcomes

Figure 3 decomposes the model-level tradeoffs. No configuration dominates; model choice should follow the deployment objective:

- **Task completion:** Grok leads every language at 75.6% but has the lowest Generation (51.8%).
- **Quality:** GPT xhigh leads Generation (67.4%) at 54.4% task completion; GPT minimal leads Interaction (45.3%) and has the lowest latency (1.12 s).
- **Stability:** Grok varies least in task completion (9.4-point range), Gemini high in Interaction (11.5), and GPT xhigh in Generation (12.3); Grok varies most in both quality scores.
- **Latency:** Higher reasoning raises response latency (GPT: 1.12–1.42 s; Gemini: 1.36–1.82 s). GPT/Gemini calls take 11.3–13.8

min, versus 25.6 min for Grok.

4.6. Validation and Robustness

Pooled held-out F1 is 96% for tool use and 88% for both naturalness and speech fidelity; the remaining Interaction components are deterministic. Among measures meeting the validation thresholds, naturalness spans Hindi (F1 81.4%, $\kappa = .634$) to Spanish (F1 97.9%, $\kappa = .965$), and speech fidelity spans Korean (F1 82.4%, $\kappa = .754$) to Spanish (F1 96.6%, $\kappa = .955$).

The separate caller audit also supports the evaluation contract. Reviewers attribute only 5/180 first critical errors (2.78%) and 5/104 failed calls (4.81%) to the simulator, and none to infrastructure. The eight caller-quality dimensions average 3.05/4; backchannel naturalness is lowest at 2.86.

Every pair of systems differs significantly on utterance Generation; the weakest contrast is Gemini minimal versus high ($p = .01844$). Interaction effect sizes preserve the substantive distinction between modest Spanish/Portuguese/Hindi degradation and large Korean/Mandarin degradation.

5. LIMITATIONS

We study six languages, three domains, three hosted model families, and one speech stack. We did not identify an open-weight model with documented support for all five localized languages, full-duplex speech, and live tool use. Evaluation costs limited additional languages, models, and trials. Simulated callers may not reproduce human timing or repair. The caller consumes agent transcripts to minimize simulator-side ASR noise, so synthesis errors cannot affect task outcomes. GPT and Gemini have two trials versus Grok’s one; validation sets are finite; and the voice-text and localization ablations change bundled conditions, so their effects remain exploratory.

6. CONCLUSION

τ -Multilingual shows that multilingual voice-agent reliability extends beyond task completion. Spanish, Portuguese, and Hindi largely match English task completion, whereas Korean and Mandarin lose ground through authentication, entity, and interaction failures. Model rankings shift with the objective, so joint reporting of task completion, interaction, and generation quality exposes failures hidden by English-only or outcome-only benchmarks. We release language packs, validated judges, and tools for community-built multilingual evaluation.

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7. COMPLIANCE WITH ETHICAL STANDARDS

The benchmark uses no real customer records or recordings. Records are synthetic and non-sensitive; calls use simulated callers and provider voices. Reviewers authored benchmark materials and annotated outputs; they consented and were compensated. Before release, artifacts were screened for incidental personal data or secrets, and flagged items were excluded.

The benchmark evaluates voice-agent systems, not people or languages; its results should not be interpreted as measures of human ability or cultural quality.

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8. REFERENCES

- [1] World Bank, “Digital progress and trends report 2025: Strengthening AI foundations,” Tech. Rep., World Bank, 2025, Accessed: Sep. 16, 2026. [Online]. Available: <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099112525160536089>.
- [2] S. Ray, K. Dhandhanian, V. Barres, and K. Narasimhan, “ τ -Voice: Benchmarking full-duplex voice agents on real-world domains,” *arXiv preprint arXiv:2603.13686*, 2026.
- [3] T. Bogavelli et al., “EVA-Bench: A new end-to-end framework for evaluating voice agents,” *arXiv preprint arXiv:2605.13841*, 2026.
- [4] G.-T. Lin, C. Chen, Z. Chen, and H.-Y. Lee, “Full-Duplex-Bench-v3: Benchmarking tool use for full-duplex voice agents under real-world disfluency,” *arXiv preprint arXiv:2604.04847*, 2026.
- [5] Sierra Research, “TAU-Multilingual codebase and caller-audit rubric,” GitHub Repository, 2026, Accessed: Sep. 16, 2026. [Online]. Codebase: <https://github.com/sierra-research/tau2-bench/tree/tau-multilingual>. Rubric: https://github.com/sierra-research/tau2-bench/blob/tau-multilingual/src/tau2/annotation/packages/assets/voice_user_simulator_rubric.json.
- [6] R. Fukuda et al., “M3-DuplexBench: A multi-turn, multilingual, multidomain benchmark for full-duplex spoken dialogue models,” *arXiv preprint arXiv:2607.29125*, 2026.
- [7] D. Jain, H. Shukla, G. Rajeev, A. Kulkarni, C. Khatri, and S. Agarwal, “VoiceAgentBench: Are voice assistants ready for agentic tasks?,” *arXiv preprint arXiv:2510.07978*, 2025.
- [8] B. Lee, I. Calapodescu, M. Gaido, M. Negri, and L. Besacier, “Speech-MASSIVE: A multilingual speech dataset for SLU and beyond,” in *Interspeech 2024*, 2024, pp. 817–821.
- [9] M. Moradshahi et al., “X-RiSAWOZ: High-quality end-to-end multilingual dialogue datasets and few-shot agents,” in *Findings of the Association for Computational Linguistics: ACL 2023*, 2023, pp. 2773–2794, Association for Computational Linguistics.
- [10] B. Ding et al., “GlobalWoZ: Globalizing MultiWoZ to develop multilingual task-oriented dialogue systems,” in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 1639–1657, Association for Computational Linguistics.
- [11] M. Freitag, G. Foster, D. Grangier, V. Ratnakar, Q. Tan, and W. Macherey, “Experts, errors, and context: A large-scale study of human evaluation for machine translation,” *Transactions of the Association for Computational Linguistics*, vol. 9, pp. 1460–1474, 2021.
- [12] C. Whitehouse et al., “MENLO: From preferences to proficiency—evaluating and modeling native-like quality across 47 languages,” in *International Conference on Learning Representations*, 2026.
- [13] S. Yao, N. Shinn, P. Razavi, and K. R. Narasimhan, “ τ -bench: A benchmark for tool-agent-user interaction in real-world domains,” in *The Thirteenth International Conference on Learning Representations*, 2025.
- [14] V. Barres, H. Dong, S. Ray, X. Si, and K. Narasimhan, “ τ^2 -Bench: Evaluating conversational agents in a dual-control environment,” *arXiv preprint arXiv:2506.07982*, 2025.
- [15] OpenAI, “GPT-Realtime-2 model,” OpenAI API Documentation, 2026, Accessed: Sep. 11, 2026. [Online]. Available: <https://developers.openai.com/api/docs/models/gpt-realtime-2>.
- [16] Google DeepMind, “Gemini 3.1 Flash Live Preview,” Gemini API Documentation, 2026, Accessed: Sep. 11, 2026. [Online]. Available: <https://ai.google.dev/gemini-api/docs/models/gemini-3.1-flash-live-preview>.
- [17] xAI, “Grok Voice Think Fast 1.0,” xAI News, 2026, Accessed: Sep. 11, 2026. [Online]. Available: <https://x.ai/news/grok-voice-think-fast-1>.
- [18] OpenAI, “GPT-5.5 model,” OpenAI API Documentation, 2026, Accessed: Sep. 11, 2026. [Online]. Available: <https://developers.openai.com/api/docs/models/gpt-5.5>.
- [19] ElevenLabs, “Eleven v3,” ElevenLabs Documentation, 2026, Accessed: Sep. 11, 2026. [Online]. Available: <https://elevenlabs.io/docs/overview/models>.
- [20] Google DeepMind, “Gemini 3.1 Pro Preview,” Gemini API Documentation, 2026, Accessed: Sep. 11, 2026. [Online]. Available: <https://ai.google.dev/gemini-api/docs/models/gemini-3.1-pro-preview>.